AI Chest X-Ray Triage Tool: Clinical Safety and Performance Assurance

Independent assurance report with ISO 14971 hazard analysis. Draft for management response.

1. Document control

Field	Detail
Report reference	AIG-2026-002
Version	1.0 DRAFT
Date	2 November 2026
Prepared by	AI Governance / GRC analyst
Reviewed by	Pending (Clinical Safety Officer)
Distribution	CMIO, Radiology lead, Clinical Safety Officer, DPO, Procurement
Classification	Confidential. Fictional engagement; synthetic data.
Engagement type	Pre-deployment clinical AI assurance
Next review	On vendor questionnaire return, or 2026-12-15

2. Executive summary

> **Overall opinion: Unsatisfactory**
> The tool does not meet the pre-agreed sensitivity minimum overall or in eight of nine subgroups, its scores are not calibrated, and the mitigation most often proposed (threshold re-tuning) does not work on held-out data. Triage use is not approved.

We assessed a vendor AI tool that prioritises adult chest X-rays on a radiology worklist, using 8,000 synthetic cases from three imaging settings (prevalence 8.7%). Although it only reorders the worklist, a low score delays reading, so a false negative is a patient-safety event.

Headline results: sensitivity 73.5% (minimum 85%), specificity 88.7%, PPV 38.3%, AUC 0.897. Aggregate AUC looks respectable. The averages hide wide differences: 87% at the validated site against 57% on the old scanner, 66% at the rural clinic and 48% in patients aged 80 and over, the group with the highest disease prevalence.

Scores overstate risk (mean predicted 30% against 8.7% observed). We tested the obvious fix, lowering the threshold per site, using a tune-on-half, test-on-half design. It does not reach 85% on held-out data and roughly doubles alert volume. The weakness is in discrimination at those sites, not the cut-off.

We recommend a time-limited second-reader pilot at the validated site only, with every study read by a radiologist. Eight governance controls remain untested pending the vendor questionnaire.

Key measure	Result	Benchmark / criterion
Overall sensitivity	73.5% (CI 70.1 to 76.7)	>= 85%
Worst subgroup sensitivity	48.3% (age 80 plus)	>= 85%
Sensitivity, old scanner site	56.6%	>= 85%
Positive predictive value	38.3%	Workload-dependent
Calibration slope / ECE	1.81 / 21.7 pts	1.0 / under 5 pts
Estimated missed cases per year (60k studies)	1,380	As low as reasonably practicable
Controls untested	8 of 14	0

Decision requested

- Confirm triage use is not approved; approve a second-reader pilot at Hospital A only (Action 1).
- Approve issue of the questionnaire and opening of the hazard log (Actions 2 and 3).
- Approve removal of numeric scores from the display (Action 4).

3. Objective and scope

To provide independent assurance on whether the AI chest X-ray triage tool performs well enough, equitably and safely for use in our imaging services, to analyse the clinical hazards it introduces, and to recommend conditions for use.

In scope	Out of scope
Sensitivity, specificity, PPV, AUC and calibration	Image-level clinical review
Subgroups: sex, age band, imaging site	Vendor software code review
Operating-point (threshold) options	Prospective clinical trial
Hazard analysis to ISO 14971	Procurement pricing
Governance evidence via questionnaire	

4. Methodology and criteria

Structure: NIST AI RMF (GOVERN, MAP, MEASURE, MANAGE). Risk process: ISO 14971 (hazard identification, risk estimation, evaluation, control, residual risk, verification). Software category: IMDRF SaMD framework, assessed as Category II (informs clinical management of a serious condition), rising to Category III if the tool were used to rule out disease. Quality of evidence assessed against TRIPOD+AI and STARD reporting expectations.

Statistics: confusion-matrix metrics with Wilson 95% confidence intervals; Fisher exact tests of sensitivity between subgroups; calibration slope, expected calibration error and Brier score; threshold analysis with a tune-on-half, test-on-half holdout; annual impact projected for 60,000 studies.

Governance controls were assessed through a separate vendor questionnaire (Excel workbook, 30 questions). Findings use condition, criteria, cause, effect and recommendation. Risk uses a 5 by 5 severity by probability matrix.

Criteria source	How it was used
ISO 14971:2019	Hazard analysis, risk acceptability, verification of controls
IMDRF SaMD N41 / N12	Clinical evaluation and risk categorisation
EU MDR and EU AI Act (Arts 9 to 15, 72)	High-risk medical AI obligations
FDA SaMD / AI-enabled device guidance and PCCP	Change control for learning devices
DCB0129 / DCB0160 or equivalent	Clinical safety case and hazard log
IEC 62304, IEC 62366-1, IEC 81001-5-1	Software lifecycle, usability, health software security
NIST AI RMF 1.0; TRIPOD+AI; WHO AI for health ethics	Measurement and reporting practice

Assurance rating scale

Rating	Meaning
Satisfactory	Safe for intended use with standard controls
Needs improvement	Use possible with specific safety conditions
Unsatisfactory	Use should not proceed (or stay limited) until remediated

Limitations and assumptions

- All data is synthetic; the model behaviour is simulated. No clinician reviewed images.
- Sensitivity minimum of 85% is an assumption for this exercise; a real minimum is set by clinical consensus.
- Threshold tuning used a single random split; a real study would use cross-validation or a separate cohort.
- Controls C-07 to C-14 are untested pending vendor evidence.

5. Results of control testing

Each control objective was tested against the stated criteria. Results are PASS, PARTIAL or FAIL, with the test performed and the evidence relied on.

Ref	Control objective	Test performed	Result	Evidence / finding
C-01	Overall sensitivity at least 85%	Confusion matrix at vendor threshold, Wilson CI	**FAIL**	73.5% (CI 70.1 to 76.7)
C-02	Sensitivity at least 85% in each subgroup	9 subgroups (sex, age, site)	**FAIL**	8 of 9 below
C-03	No clinically important disparity across settings	Fisher exact tests vs reference	**FAIL**	Hospital B $p < 0.001$; age 80 plus $p < 0.001$
C-04	Scores calibrated if shown as risk	Calibration slope, ECE, Brier	**FAIL**	ECE 21.7 pts, slope 1.81
C-05	Alert burden acceptable	PPV, alert rate, annual volume	**PARTIAL**	PPV 38.3%, ~9,998 flags/yr
C-06	Local re-tuning can meet the minimum	Tune-on-half, test-on-half	**FAIL**	Held-out sensitivity 77 to 78% at weak sites
C-07	Regulatory status matches intended use (FDA / CE)	Questionnaire section 1	NOT TESTED	Awaiting vendor response (F6)
C-08	Prospective or external clinical validation	Questionnaire section 2	NOT TESTED	Awaiting vendor response (F6)
C-09	Human factors and automation-bias study	Questionnaire section 5	NOT TESTED	Awaiting vendor response (F6)
C-10	Cybersecurity and SBOM	Questionnaire section 6	NOT TESTED	Awaiting vendor response (F6)
C-11	Post-market surveillance and drift monitoring	Questionnaire section 7	NOT TESTED	Awaiting vendor response (F6)
C-12	Change control and predetermined change plan	Questionnaire section 7	NOT TESTED	Awaiting vendor response (F6)
C-13	Data protection, retention and secondary use	Questionnaire section 6	NOT TESTED	Awaiting vendor response (F6)
C-14	Clinical safety case and hazard log ownership	Questionnaire section 5	NOT TESTED	Awaiting vendor response (F6)

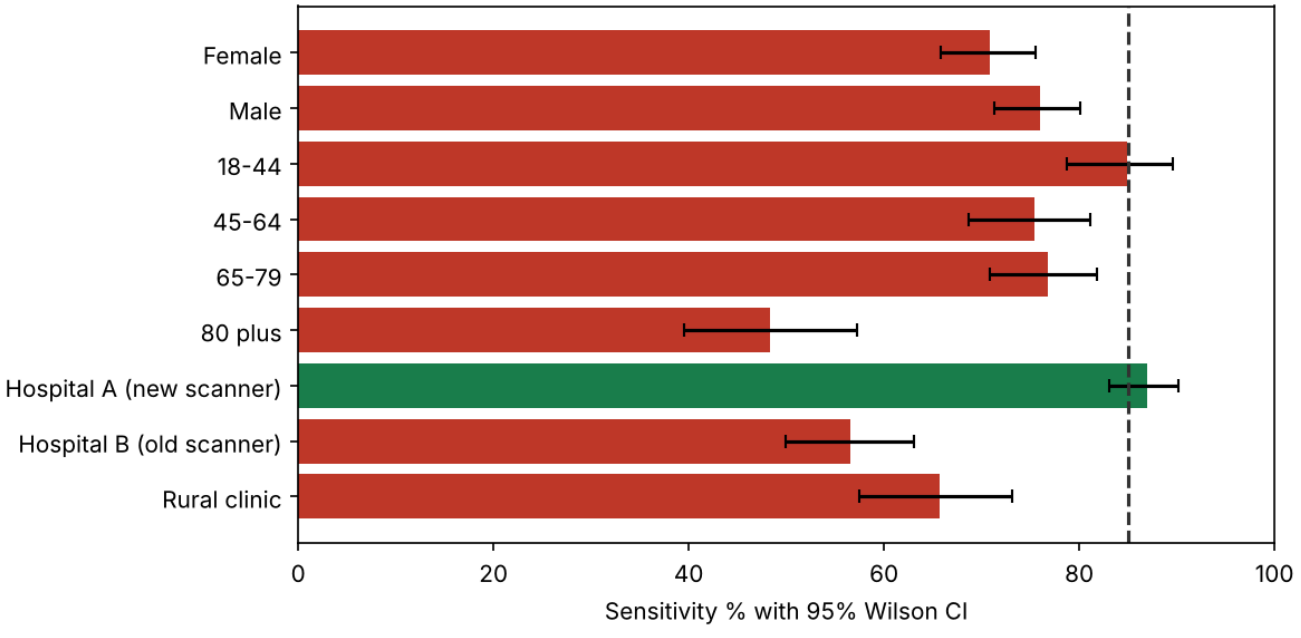


Figure 1. Sensitivity by subgroup with 95% Wilson confidence intervals (dashed line: 85% minimum)

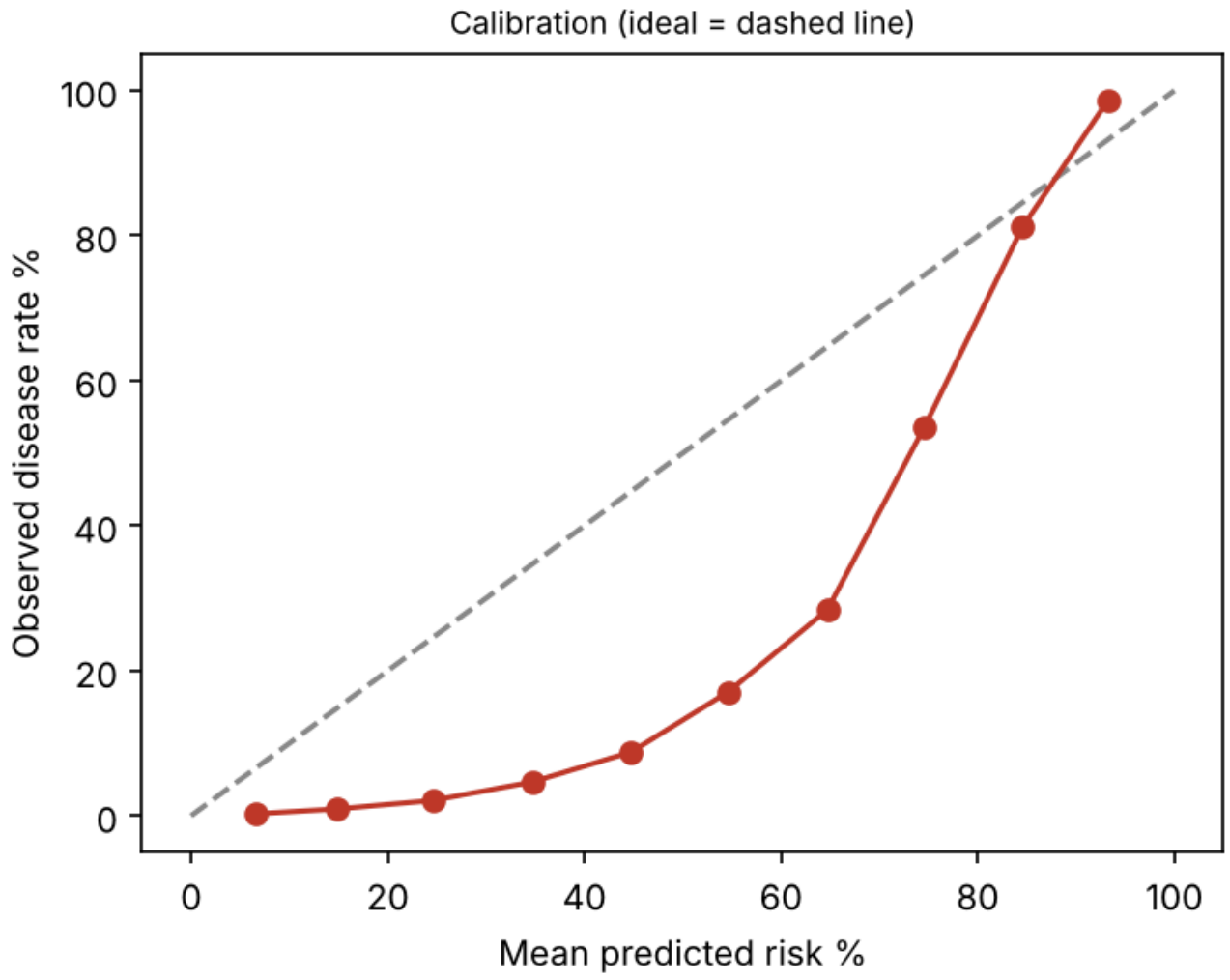


Figure 2. Calibration: predicted versus observed disease rate

6. Detailed findings

F1 Sensitivity is below the 85% minimum overall and in most subgroups [Critical]

Field	Detail
Rating	**Critical**
Condition (what we found)	At the vendor's threshold (0.5), overall sensitivity is 73.5% (95% CI 70.1 to 76.7); 8 of 9 subgroups are below 85%. At 60,000 studies a year this equates to about 1,380 true cases not flagged.
Criteria (what should be)	Pre-agreed clinical performance minimum of 85% sensitivity (set by the clinical safety officer for a triage tool where a missed case delays reading). IMDRF SaMD clinical evaluation expects performance in the intended population.
Cause	Operating point chosen for specificity and demonstrated on a single favourable site; no subgroup performance requirement specified in procurement.
Effect (risk)	Delayed diagnosis and potential serious patient harm, concentrated in the groups below.
Recommendation	Do not approve for triage. Restrict any pilot to a second-reader role (all studies still read) until local performance meets the minimum.
Framework references	ISO 14971 7; IMDRF SaMD N41; EU MDR Annex I; EU AI Act Art 15
Owner / target	Clinical Safety Officer / 2026-11-16

F2 Performance fails on the older scanner, in rural imaging and in the oldest patients [Critical]

Field	Detail
Rating	**Critical**

Field	Detail
Condition (what we found)	Sensitivity: Hospital A 87.0%; Hospital B (old scanner) 56.6% (p < 0.001 vs A); rural clinic 65.7% (p < 0.001); age 80 plus 48.3% versus 84.9% for 18 to 44 (p < 0.001). Sex difference (70.9% v 76.0%) is not significant (p = 0.144).
Criteria (what should be)	Performance must be demonstrated and acceptable in each clinically relevant subgroup and imaging setting in the intended use population (EU AI Act Art 10(2) and 15; MDR clinical evaluation; WHO guidance on AI for health).
Cause	Dataset shift: the model degrades on image characteristics unlike its training data (scanner, acquisition protocol). The 80 plus group also has the highest disease prevalence, so the tool is weakest where need is greatest.
Effect (risk)	Inequity of care: the sites and patients with least radiology capacity receive the least reliable support.
Recommendation	Restrict use to the validated scanner class. Require vendor to provide subgroup validation and a fine-tuning plan using local data for other sites, followed by re-validation.
Framework references	NIST AI RMF MEASURE 2.11, MAP 2.3; ISO 14971 5.5
Owner / target	CMIO / Vendor / 2026-12-15

F3 Risk scores are not probabilities and must not be displayed as percentages [High]

Field	Detail
Rating	**High**
Condition (what we found)	Mean predicted risk is 30.3% against an observed prevalence of 8.7%. Expected calibration error 21.7 points; calibration slope 1.81 (ideal 1.0); Brier score 0.109. A score band of 0.4 to 0.6 corresponds to about 9% observed disease.
Criteria (what should be)	Where a probability is displayed it must be calibrated (TRIPOD+AI reporting; NIST MEASURE 2.5).
Cause	Model trained for ranking, not calibrated to deployment prevalence.
Effect (risk)	Clinicians reading a number as a probability will over-estimate risk in most cases and mis-prioritise.
Recommendation	Show a coarse flag only; or recalibrate on local data and re-validate before displaying any numeric risk.
Framework references	TRIPOD+AI; NIST AI RMF MEASURE 2.5
Owner / target	Vendor / CMIO / 2026-12-15

F4 Re-tuning the threshold per site does not restore safe performance [High]

Field	Detail
Rating	**High**
Condition (what we found)	Thresholds were tuned on half of each group's data to reach 85% sensitivity and evaluated on the other half. Result on held-out data: Hospital B (old scanner): 78% sensitivity at threshold 0.33, PPV 18%, alerts on 35% of studies; Rural clinic: 78% sensitivity at threshold 0.38, PPV 24%, alerts on 30% of studies; Age 80 plus: 77% sensitivity at threshold 0.26, PPV 20%, alerts on 53% of studies. None reach 85% on held-out data, and alert volume rises sharply.
Criteria (what should be)	Mitigations must be shown to be effective on data not used to design them (ISO 14971 7.2 verification of risk control effectiveness).
Cause	The problem is weak discrimination (AUC) at those sites, not the cut-off. Lowering the threshold buys sensitivity with false positives, not with better signal.
Effect (risk)	A tempting quick fix would add workload and still leave patients under-served.
Recommendation	Do not rely on threshold changes. Require fine-tuning on local images with prospective re-validation.
Framework references	ISO 14971 7.2; IMDRF SaMD N41
Owner / target	Vendor / Radiology lead / 2027-01-31

F5 Alert burden is high and most alerts are false positives [Medium]

Field	Detail
Rating	**Medium**
Condition (what we found)	At prevalence 8.7% the PPV is 38.3% and 16.7% of studies are flagged (about 9,998 per year at 60,000 studies).
Criteria (what should be)	Workflow impact is assessed and acceptable (human factors, IEC 62366-1).
Cause	Moderate specificity at low prevalence.
Effect (risk)	Alert fatigue, ignored flags, slower worklist.
Recommendation	Pilot with measured workload and override rates; define tolerance and stop rules.
Framework references	IEC 62366-1; NIST MEASURE 2.8
Owner / target	Radiology lead / 2027-02-28

F6 Clinical safety governance evidence is missing [High]

Field	Detail
Rating	**High**
Condition (what we found)	Eight control objectives (C-07 to C-14) are untested: no regulatory clearance mapping, no prospective validation, no human factors study, no cybersecurity assessment, no post-market surveillance plan, no change-control (PCCP) and no hazard log owner.
Criteria (what should be)	DCB0129/DCB0160 (UK) or equivalent clinical safety standard; MDR; EU AI Act Arts 9, 14, 72; FDA predetermined change control guidance.
Cause	Procurement proceeded on a vendor demonstration.
Effect (risk)	Unquantified patient-safety and regulatory exposure.
Recommendation	Vendor to return the clinical AI due-diligence questionnaire (workbook provided); clinical safety officer to maintain the hazard log and safety case.
Framework references	ISO 14971; IEC 62304; IEC 81001-5-1
Owner / target	Clinical Safety Officer / 2026-11-30

7. Risk assessment

Hazards follow ISO 14971: impact is harm severity (1 negligible to 5 catastrophic) and likelihood is probability of the harm occurring (1 improbable to 5 frequent). Inherent assumes no controls; residual assumes the proposed controls are implemented and verified.

ID	Risk	Inherent	Existing controls	Residual	Owner	Treatment
H1	False negative deprioritises a true case; reading delayed	4x5=20 (Critical)	None at present	2x5=10 (Medium)	Clinical Safety Officer	Read-all-studies policy; restrict to validated scanner; sample audit of unflagged
H2	Radiologist over-trusts a negative flag (automation bias)	3x5=15 (High)	None at present	2x5=10 (Medium)	Radiology lead	Training; blinded-read audits; no 'normal' label shown
H3	Scanner or protocol upgrade silently changes performance	4x4=16 (High)	None at present	2x4=8 (Medium)	Imaging physicist	Change log from imaging team; revalidate before use on new equipment
H4	Numeric score misread as probability	4x4=16 (High)	None at present	1x4=4 (Low)	CMIO	Remove numeric display; coarse flag only
H5	Inequitable performance in 80 plus and rural patients	4x4=16 (High)	None at present	2x4=8 (Medium)	CMIO	Scope exclusion until validated; subgroup monitoring
H6	Alert fatigue leads to ignored flags	4x3=12 (High)	None at present	3x3=9 (Medium)	Radiology lead	Workload metrics; override-rate stop rules
H7	Unvalidated vendor model update deployed	3x4=12 (High)	None at present	1x4=4 (Low)	Procurement	Contractual change notice; revalidation gate
H8	Image data used for vendor training without basis	3x4=12 (High)	None at present	1x4=4 (Low)	DPO	DPA prohibiting secondary use; DPIA

8. Recommendations and action plan

#	Action	Addresses	Priority	Owner	Target date
1	Record decision: no triage use; second-reader pilot at Hospital A only	F1 F2	Immediate	Clinical Safety Officer	2026-11-09
2	Issue clinical AI due-diligence questionnaire; collect evidence	F6	**High**	Procurement / CMIO	2026-11-30
3	Open hazard log and clinical safety case; name hazard owners	F6	**High**	Clinical Safety Officer	2026-11-30
4	Remove numeric score display; show coarse flag	F3	**High**	CMIO	2026-12-15
5	Vendor to supply subgroup validation and a local fine-tuning plan; re-validate prospectively	F2 F4	**High**	Vendor / Radiology lead	2027-01-31
6	Run pilot with read-all policy, 5% blinded re-read of unflagged studies, workload and override metrics	F1 F5	**Medium**	Radiology lead	2027-02-28
7	Embed monitoring plan with stop rules (appendix B) in contract and local procedure	H3 H6 H7	**Medium**	Clinical Safety Officer	2027-03-31

9. Management response

Responses below are drafted for the accountable owner to confirm, amend or reject. An accepted risk requires a named executive signature and an expiry date.

Finding	Proposed response	Accept / Amend / Reject	Owner sign-off
F1	Agree. Triage use not approved.	Pending	Pending
F2	Agree. Scope restricted to Hospital A.	Pending	Pending
F3	Agree. Numeric display disabled.	Pending	Pending
F4	Agree. Threshold re-tuning removed from the mitigation plan.	Pending	Pending
F5	Agree.	Pending	Pending
F6	Agree.	Pending	Pending

10. Appendices

Appendix A. Subgroup performance

Subgroup	N	Sens %	95% CI	Spec %	PPV %	AUC
sex: Female	4027	70.9	65.8 to 75.5	89.2	37.2	0.883
sex: Male	3973	76.0	71.3 to 80.1	88.3	39.3	0.910
age_band: 18-44	2440	84.9	78.7 to 89.6	89.3	36.6	0.937
age_band: 45-64	2823	75.4	68.7 to 81.1	88.1	30.5	0.904
age_band: 65-79	1926	76.8	70.9 to 81.8	89.3	49.0	0.927
age_band: 80 plus	811	48.3	39.5 to 57.2	88.2	41.0	0.762
site: Hospital A (new scanner)	4086	87.0	83.0 to 90.1	88.4	40.9	0.947
site: Hospital B (old scanner)	2397	56.6	49.9 to 63.1	88.7	32.7	0.826
site: Rural clinic	1517	65.7	57.4 to 73.1	89.9	39.1	0.883

Appendix B. Monitoring plan and stop rules

Signal	Method	Frequency	Stop trigger
Sensitivity	Blinded re-read of 5% of unflagged studies	Monthly	Estimate below 85%
Subgroup sensitivity	Stratified by site, age, sex	Quarterly	Any subgroup below 80%
Alert rate	Share of studies flagged	Weekly	Moves more than 20% from baseline
PPV	Flags confirmed	Monthly	Below 25%
Override rate	Radiologist disagrees	Monthly	Over 40% or near 0%
Equipment change	Imaging change log	Event	Revalidate before use

Appendix C. Threshold re-tuning on held-out data

Group	Threshold	Held-out sens at 0.5	Held-out sens tuned	Tuned PPV	Tuned alert rate
Hospital A (new scanner)	0.50	88.9	88.9	40.0	18.0
Hospital B (old scanner)	0.33	49.5	78.2	18.2	35.5
Rural clinic	0.38	69.1	77.9	23.9	29.9
Age 80 plus	0.26	36.8	77.2	20.1	53.2

Appendix D. Companion documents

docs/clinical_ai_vendor_due_diligence_questionnaire.xlsx; data/cases.csv; analysis.py (simulation) and audit.py (statistics); data/clinical_stats.json.